

Perceptions of Mental Health Crisis among U.S. Military Veteran Peer Mentors and Potential of Mobile-Based Peer-Support Interventions

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ABSTRACT

U.S. military veterans face an elevated risk of suicide or exhibiting suicidal behaviors. Peer-support interventions have proven to be effective for veterans because of their shared experience and community. However, as veterans might face a mental health crisis at any time, it is crucial that the mentors can identify their peers' crisis warning signs early enough. Mobile technology has the potential to facilitate and improve peer-to-peer communication. Gaining an understanding of how the veteran community perceives crisis symptoms, as well as their values and technological needs, bears utmost importance in creating any tool or adopting any strategy. Hence, we conducted a mixed-methods study with twelve peer mentor military veterans. Our research offers an in-depth understanding of the nuanced conceptions of mentor veterans about early warning signals and acute mental health crisis symptoms, as well as aspects of technology that may aid this community in recognizing and managing these symptoms.

CCS CONCEPTS

• **Human-centered computing** → *Empirical studies in HCI*; **Empirical studies in ubiquitous and mobile computing**.

KEYWORDS

U.S. military veteran, Mental health, Perceptions of crisis, Peer-support, Focus group.

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1 INTRODUCTION

One in five veterans experiences mental health (MH) concerns, including post-traumatic stress disorder (PTSD), major depression, and anxiety with the increased risk of unhealthy alcohol and drug use, substance use disorders, and suicidal behavior. [15]. Differences in age-adjusted rates of suicide between 2005 and 2015 were greater for U.S. Veterans than non-Veterans (32.7% vs 19.9%, respectively) [7]. Peer support interventions have been proven effective to improve veteran mental health in the past [8, 11, 12]. A peer mentor refers to a veteran who has dealt with difficulties of the re-adjustment to the civilian life phase. Peer mentors can help better in mentees' transition to civilian life if expectations are well-understood and mental health-related crises are anticipated early [3]. Therefore, peer-support-based interventions usually emphasize building and maintaining continuous mentor-mentee communication, so that mentors can identify any potential symptoms of crisis and take action [24]. A mental health crisis is any circumstance in which a person's actions put them at risk of hurting themselves or others and/or making it impossible for them to take care of themselves or carry out their daily responsibilities [2]. However, the extent to which available peer-support intervention programs for veterans successfully employ crisis detection and prevention strategies, and their effectiveness remains largely unknown [9]. Moreover, even with different interventions in place, many veterans with PTSD find it difficult to share their experiences and mental



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health conditions with mentors [9, 10, 24]. As a result, premature treatment dropout and poor compliance are common challenges in treating veterans through peer support [12]. Providing a systematic way to improve the quality of communication between a peer mentor and their mentees may be promising. To improve the quality of veteran peer support, it is crucial to understand the nuanced conceptions of mental health crises by the mentors, as these perceptions and how they respond to them can affect the detection and prevention of their peers' crises-related symptoms and outcomes. In this article, we aim to answer the following research questions:

RQ1: What are the symptoms of crises among U.S. War Veterans as observed by their peer mentors?

RQ2: What technological interventions can be implemented to increase treatment adherence and to facilitate the mentor-mentee communication among veterans?

We followed a mixed-methods approach, combining data from survey responses and focus groups. We collaborated with Medical College of Wisconsin and a non-profit veteran-led organization - "Dryhooch of America". CSCW and HCI researchers have highlighted the importance of designing mental health technologies for and with peer supporters to make the peer support tools empowering, accessible, and safe [20, 29, 29]. Veterans are a unique marginalized group facing multiple vulnerabilities. They experience trauma differently than civilians do, and these events affect how they seek and receive peer support [24]. To create solutions that will work to meet the unmet mental health needs of this particular community, it is essential to understand how they perceive and deal with mental health crises. We contribute to the existing literature on peer support technologies for mental health by discussing the design implications of technological tools for peer mentors to detect potential crisis moments of their mentee veterans.

2 RELATED WORK

Peer support interventions for veterans: A variety of treatments have been widely disseminated and promoted throughout both the Veteran Affairs (VA) and other veteran-led community organizations [18]. Mobile apps and websites have been developed recently, particularly to provide peer-support. For example, a web-based peer-support based problem solving system (Moving Forward) has been proved successful in decreasing depression symptoms in veterans [21]. A smartphone application, QRF, was developed to collect basic self-report information using an Ecological Momentary Assessment (EMA) approach, and the peer mentors could visualize the self-report data to identify mental health conditions and make informed decisions about checking in with their mentees [24, 25].

When veterans leave the military and re-adjust to civilian life, they feel disconnected from their social groups [27]. To regain a sense of control, oftentimes they reach out to fellow veterans on social media or peer-support groups [10]. These groups connect veterans to a familiar culture and have notable communication features that include disclosure strategies and judiciously sharing. There are now more visits to these groups each year than to mental health professionals [8].

However, adherence to the intervention programs and completion among veterans is often poor, resulting in diminished impact

[4, 22]. On one hand, there is a certain cognitive workload the mentors face to keep a healthy and constant communication with their mentees, on the other hand, the mentees feel uncomfortable sharing all their difficulties and challenges with the mentors [14, 28]. Understanding the peer support process from a mental health crisis standpoint may help both increase adherence and improve the quality of peer support, as well as guide the design of strategies and technologies in mental health related crisis detection and prevention.

Crisis management in peer-peer support: There is a need to better understand how the widespread use of peer-support interventions will transform crisis management [6, 17]. Additionally, there have been reported occasions where people and organizations have failed to recognize the possible early symptoms of a crisis [26]. Prior research pointed out that even if a potential crisis is detected and comprehended, it still may not be dealt with in the intended way [5]. Moreover, social aspects like fostering trust, establishing credibility, raising awareness, expanding knowledge of challenging information, reaching consensus on solutions, and inspiring actions are closely related to warning response. [5]. Further conceptual elaboration will be very useful for examining closely the peer response phenomenon considering previous sociological findings on emergent response during emergencies. By contrast, understanding the veteran community will require a different unit of analysis and methodological approach [10], in part because of the mixed demographic characteristics and unique context, the diversity of spaces and activities that are found in these settings, and the growing reliance on mobile phones and other new media as primary means of communication.

3 DATA COLLECTION AND ANALYSIS

We considered the determinants of mental health used in an existing commercially available peer-support-based app designed with and for veterans (QRF) [9, 23, 24]. There are two versions of the app. The veteran version collects basic self-report Ecological Momentary Assessment (EMA) responses about changes in physical health, sleep, risky behavior, stress, and self-worth [9]. The peer mentor version visualizes self-report data to assist peer mentors in identifying mental health conditions. We followed three steps to collect and analyze data:

Identifying potential scenarios of warning signs and acute crises: We conducted multiple discussion sessions with two subject matter experts: an experienced peer mentor veteran and a professional psychologist. They came up with different scenarios that may be considered as 'early warning signs' or 'acute crises' in terms of weekly change in the EMA responses. We finally considered four example scenarios representing potential early warning signs and four example scenarios representing potential acute crisis were considered.

Conducting surveys with peer mentors: The mock-up screens with the example scenarios were sent to twelve peer mentors through a web survey. The participants' age ranged between 25 and 69. They worked in different branches of the military including army, navy, marine corps, active duty, and reserves with highest in army. Their mentoring experiences varied from 0 years to 10 years. For each scenario, the participants were asked to select from one of

the four possible actions they would take: meet, call, text, or wait and see. Since “call” and “meet” are comparatively more direct and immediate action, if a peer mentor selects one of these two options, we assume that they considered that particular scenario as “acute crisis”. If “text” is selected, this scenario is considered as an “early warning sign”, and if “wait and see” is selected then the mentor considers this scenario as “no crisis”.

Conducting focus groups with peer mentors: Since there was not as much agreement in the survey responses as expected, we wanted to gain a deeper understanding of the nuances in mentors’ decision making in responding to their mentees’ warning signs [16]. We conducted several sessions of focus group to ask questions on the context and reasoning behind their responses. Study sessions were audio recorded with participants’ consent. The study was approved by the ethics review board of the authors’ institution.

The audio recordings were transcribed and anonymized by two authors. The information was then arranged by questions and themes [14, 19]. By identifying the key concepts that recurred during your focus group conversations, the main themes and ideas in the context of our study questions were interpreted.

4 FINDINGS

4.1 Survey results: little agreement between mentors in perceptions of crisis symptoms and actions

According to the subject matter experts’ opinions, we had expected that most of the mentors will indicate the examples in Fig. 1(a) as early warning signs of crisis and the examples in Fig. 1(b) as signs for acute crisis. However, from the survey responses (see corresponding tables in Fig. 1), it is clear that the veterans did not agree on all the scenarios. Particularly, for veteran A and C’s scenarios, 50% and 33.3% participants respectively marked them as ‘no crisis’ and indicated that they would not take any action if they observe these symptom changes in their mentees’ responses. The examples of veteran B and D in Fig. 1(a) do not show 100% agreement in perceived actions, either. Similarly, in Fig. 1 (b), except for the example of veteran D, more than 50% of the veterans considered the scenarios as ‘early warning signs’, and what is worse is, 2 and 1 participants respectively considered examples A and B to be a ‘no crisis’ situation. It is clear from the findings that the change in health related symptoms are perceived by different mentors differently, which can be very different from a subject matter expert’s opinion, and that also influence their immediate actions to check in with their mentees. This can have serious consequences for the mentees facing an acute crisis.

4.2 Findings from the focus groups

4.2.1 Peer mentors’ different perceptions of early warning signs and acute crisis. The early warning stage is when the crisis event has not occurred yet and the symptom changes are usually subtle, making them difficult to recognize. **Changes in daily routines and lifestyles** is perceived to be a vital early warning sign. Significant decline in regular positive and healthy behavior and activities (e.g., not answering and returning calls, sleeping more than usual, etc.) can be an indicator that something wrong is going on in their life.

“If that person gets angry, aggravated quickly, not eating, not sleeping properly...all that madness. That needs to be addressed. ... like I’m not even motivated to get up to do any housework.”

Acute crisis occurs when someone is on the verge of a mental health crisis or is already in a crisis situation (e.g., contemplating self-harm or suicide). Mentors mentioned that acute crisis signs can be identified through in-person contact only, as it allows them to notice changes in patterns and visual cues (e.g., speaking tones or body languages). The most prominent indicator of acute crisis as perceived by mentors is when a mentee **isolates** themselves, and suddenly disconnects from their mentors. Peer support works on the premise that the mentee would be willing to receive advice and support from their mentors and would apply them to cope up with their mental health conditions. Not caring about communicating with their mentors indicate that the veteran does not care about the consequences of their actions, and has lost trust in the judgement of others including their mentors or care coordinators. Some of the **risky behaviors** can also be prompted, such as heavy drinking, driving while being intoxicated, etc.

“You can watch the way their hands are, whether they start shaking, started fidgeting, or getting restless.”

Our participants emphasized on identifying **small changes** way early before they become serious, including violent acts, getting angry easily, leaving treatment early, relapse, etc. *“It starts with little things. If I’m not aware of, or addressing them, eventually, they keep growing, and my thinking will lead to my risky behaviors. I’ll start to feel not worthy of recovery”*. Mentors also recognize that many of these signs can be **false positives**. In any case, if they are identified early on, the mentors will be able to create a helpful atmosphere for the mentee to vent and share what they are struggling with.

“... Sometimes they’ll tell you that nothing’s wrong. It’s just a bad day. And that’s where you utilize the time to find out what’s going on with that day: do you need a couple minutes just to vent out?”

4.2.2 Implications for technology design for veteran peer support. Our focus group participants pointed out how technology can help them in identifying warning signs in their mentees. The mentors appreciated the existing tools that allow them to monitor health related changes through EMA survey responses, and help them identify who needs immediate help. However, they suggested that instead of asking directly about engagement in ‘risky behaviors’ it should be asked through **indirect or positive questions**. It can work as reverse psychology and might help them to open up.

“Have you engaged in risky behavior? That’s more direct. I don’t think you’re gonna get an honest answer”.

One of the peer mentors shared their preferences of letting the mentee take control of their actions: *“instead of telling him what he should do, I start with: what do you feel you should do? What do you think your next step should be?”* In addition to the **daily check-ins** provided by the current apps, the participants suggested some form of **visualization of data** to identify changes easily and quickly. It also helps them decide the right mode of communication and appropriate questions to ask based on mentees’ current situation. They emphasized that simple conversation starters and being persistent in communication are the keys for helping their mentees open up.

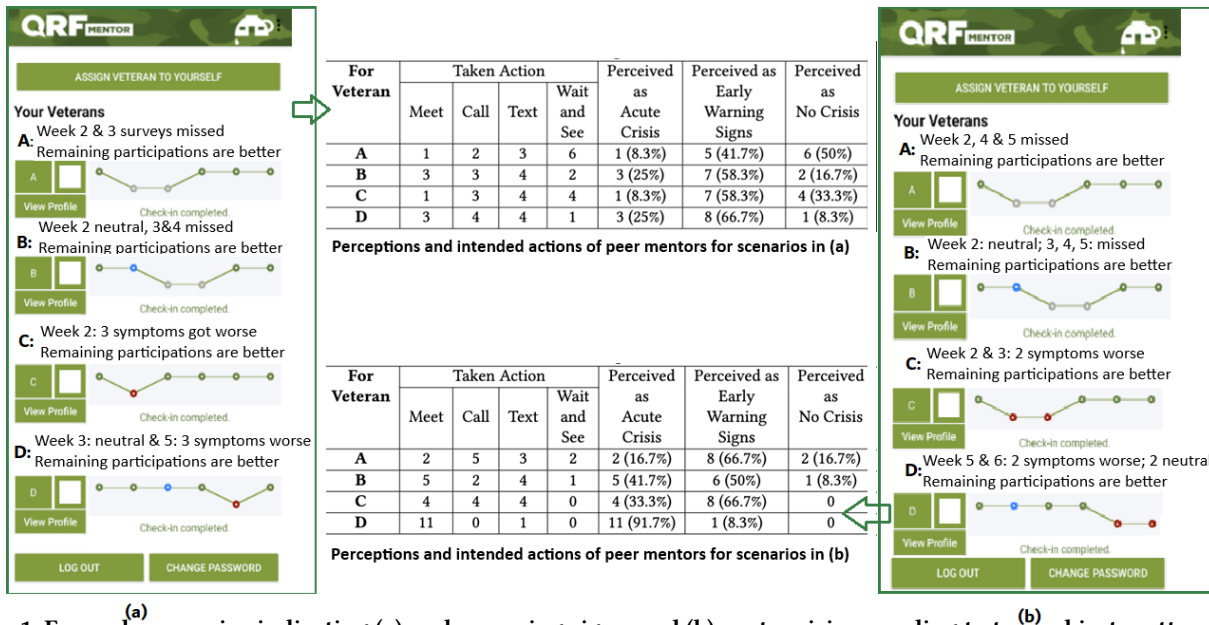


Figure 1: Example scenarios indicating (a) early warning signs, and (b) acute crisis according to two subject matter experts. The corresponding tables show how 12 peer mentors perceive each example scenario: ‘no crisis’, ‘early warning sign’, or ‘acute crisis’, and their intended actions: ‘wait and see’, ‘text’, ‘call’, or ‘meet in person’

They have also pointed out the benefits of **technology-mediated peer support for the younger generations** that tend to be less active in physical meetings. For that population of mentees, sometimes not using the app is perceived by the mentors as isolation of the mentees (e.g. “when they were doing the weekly check-ins, then suddenly **disappear** with no calls/check-ins” or “they will just ping me through the app and say, I’m still alive, and I’m still kicking”).

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However, mentors also expressed their concerns about the **exhaustion** due to the heavy toll they take from their mentees, and the need for self-care and breaks. They suggested designing technology that can share the load and keep the communication open when mentors need their space for self-care.

“If you’re not treading water to save your energy, you’re expending all energy on one individual person without setting any boundaries, you do tend to get exhausted.”

Limitations of technology: While the mentors in general appreciated the idea of designing technologies to facilitate the peer-support process, they thought pushing too much with the technology may result in serious consequences, as it might **cross the**

set boundaries and **break trust**. Moreover, text and notification alerts may result in **violating privacy** as some veterans prefer not to discuss their military careers or mental health problems to their friends and family members due to the **social stigma**. One veteran mentioned “My kid was doing social studies with my wife and they were talking about the Middle East. I’m like, Hey, I’ve been there. He looked at me, like, why was he there? I don’t talk about my military career with them.”

Additionally, some veteran might have **limited or no access to technology and the internet**. Many mentee veterans are homeless and/or unemployed, use government phones with limited minutes. So, it’s crucial to carefully examine the veteran’s environment and their access to resources while integrating technological solutions.

5 DISCUSSION

Peer mentors attempted to recognize early warning signs by observing minor changes in their mentees’ daily routines. Tracking multiple mentees’ symptoms simultaneously can be challenging. Prior research suggests the necessity of building systems to collect self-reported data using Ecological Momentary Approaches [9, 23, 24]. There are apps available with built-in symptom reporting, tracking, and monitoring [11], which can be utilized in better management of tracking multiple mentees. However, these systems frequently experience premature dropouts. To combat the low adherence, factors like **ensuring the mentor’s participation in the process and building trust between mentors and mentees** should be considered. Mentors have also stated that they viewed absence of communication as an early warning sign, and tried to get in touch with them. Visualization of self-report data or regular check-ins with proper questions identified from the mentor’s personal experience may facilitate the mentors’ intervention decisions [9].

In the context of veteran peer-support interventions, mentor veterans are recruited and trained for providing effective peer support, and in many cases, they are legally and ethically bound to provide this service. But the mentors themselves, however, can experience exhaustion and burnout in continuously follow up with their mentees to help them take a particular action needed to improve their mental health and wellbeing. Self-adherence through persuasion has the potential to help partially relieve mentors of their cognitive load. Persuasion has traditionally been regarded as a communication process in which a persuader sends a persuasive message to a persuadee with the intention of changing the recipient's attitudes or behavior [30]. Simple persuasion techniques, such as daily check-ins, gamification, reminders, and self-monitoring are generally perceived as helpful features in mHealth applications [1, 11, 13]. **persuasion techniques such as reward systems/ incentives** for regular participation will work for veterans. Military personnel are accustomed to receiving ranks and rewards for their efforts. Simulating similar conditions as rewards can encourage both peer mentors and peer mentees to pay closer attention to their commitments.

Our participants customized their mentoring based on different needs. Some prefer one-on-one time, while some others are restrictive about how they share symptoms. Some individuals are tech-savvy, while others are not. Some prefer to virtual communication, while others prefer in-person interactions. Therefore, the mentoring style needs to be matched with mentees' preferences. **Algorithmic peer matching** can be promising in this and similar health contexts.

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